



DAX

DAX Category : Filter Functions
Function Covered :

CALCULATE
CALCULATETABLE



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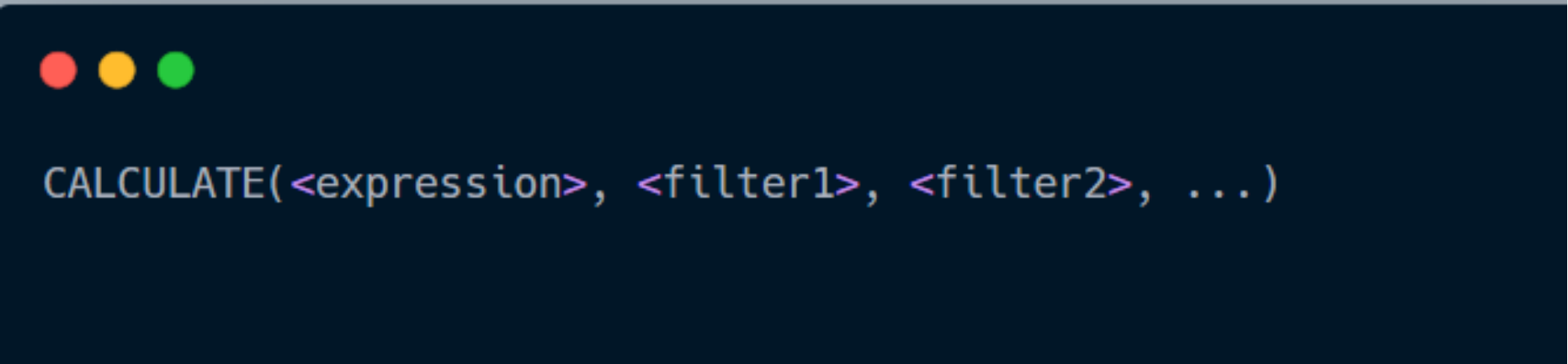


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CALCULATE Function

CALCULATE modifies the context in which a DAX expression is evaluated by applying filters to your data. It's one of the **most powerful DAX functions**, used for conditional aggregations.

Syntax



```
CALCULATE(<expression>, <filter1>, <filter2>, ...)
```

- <expression>: Any DAX expression like SUM, COUNT, etc.
- <filter>: Conditions to modify the context of calculation.

Use Cases:

1. Applying filters to specific measures: Filter sales by region, category, or date range.
2. Conditional metrics: Calculate profit for specific product categories or time periods.
3. Time Intelligence calculations: CALCULATE is essential for functions like YTD (Year-to-Date), MTD (Month-to-Date), etc.

Examples

Scenario:

You want to calculate the total sales for the "Electronics" category in 2022.



```
CALCULATE(  
    SUM(Sales),  
    Product[Category] = "Electronics",  
    YEAR([Date]) = 2022  
)
```

Scenario:

You want to calculate the total sales for products with sales greater than 200 in 2022.



```
CALCULATE(  
    SUM(Sales),  
    FILTER(  
        SalesTable,  
        SalesTable[Sales] > 200 && YEAR(SalesTable[Date]) = 2022  
    )  
)
```

CALCULATETABLE Function

CALCULATETABLE works like CALCULATE but returns a table instead of a scalar value. It filters the table based on certain conditions.

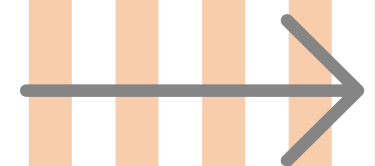
Syntax

```
CALCULATETABLE(<table>, <filter1>, <filter2>, ...)
```

- <table>: The table to be filtered.
- <filter>: The conditions used to filter the data.

Use Cases:

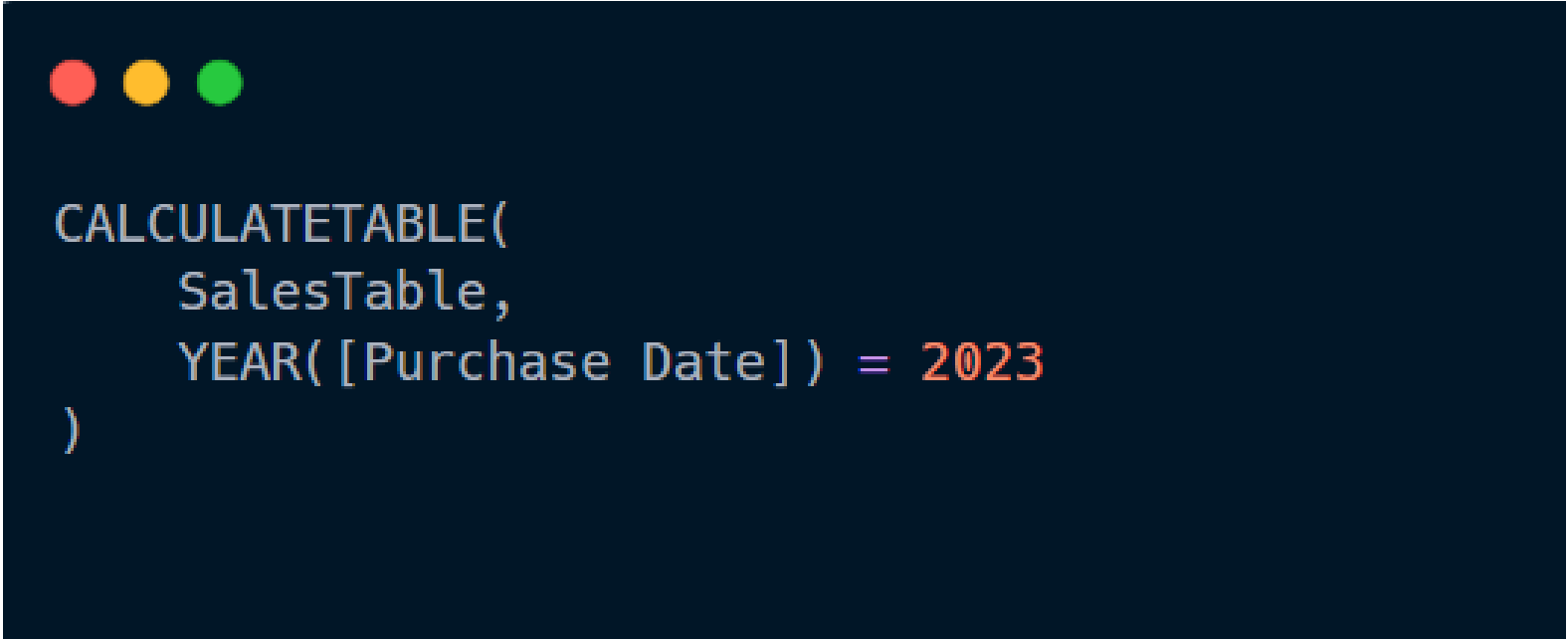
1. **Filtered tables for visualization:** Use CALCULATETABLE to display only certain subsets of data, e.g., sales from a specific region.
2. **More complex queries:** Combine multiple conditions on large datasets for refined table results.



Examples

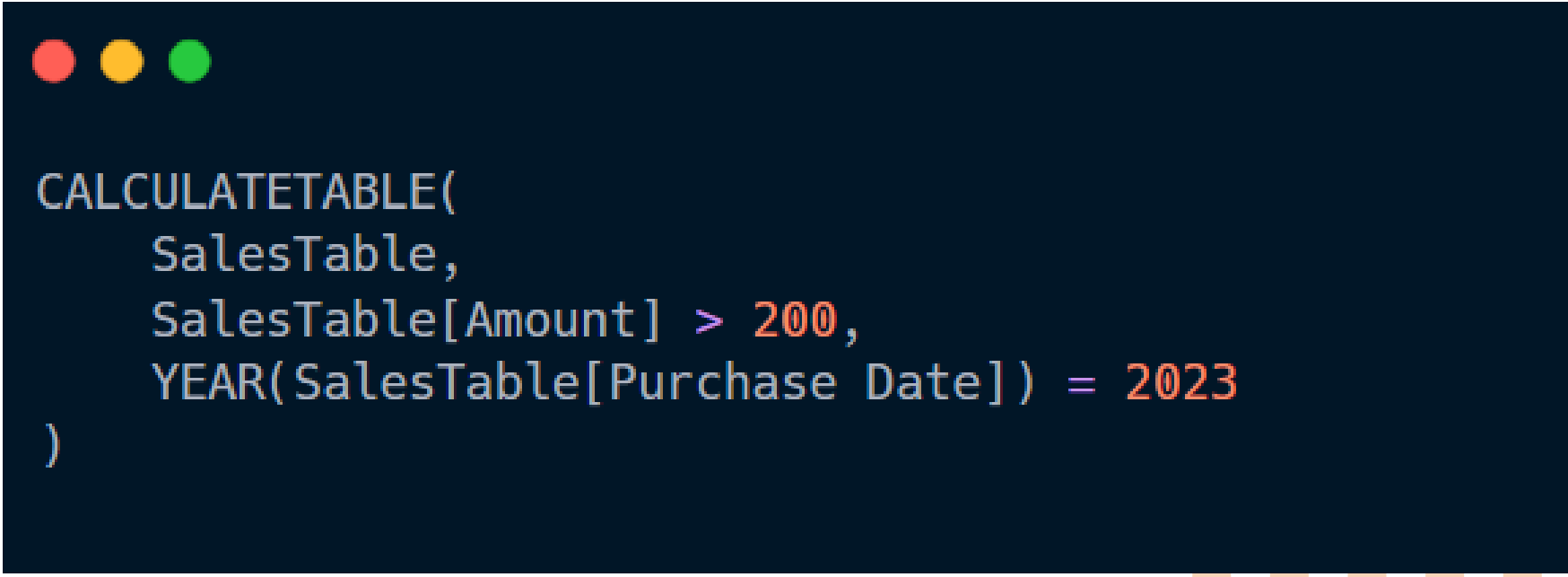
Scenario:

You want a filtered table for customers who made purchases in 2023.

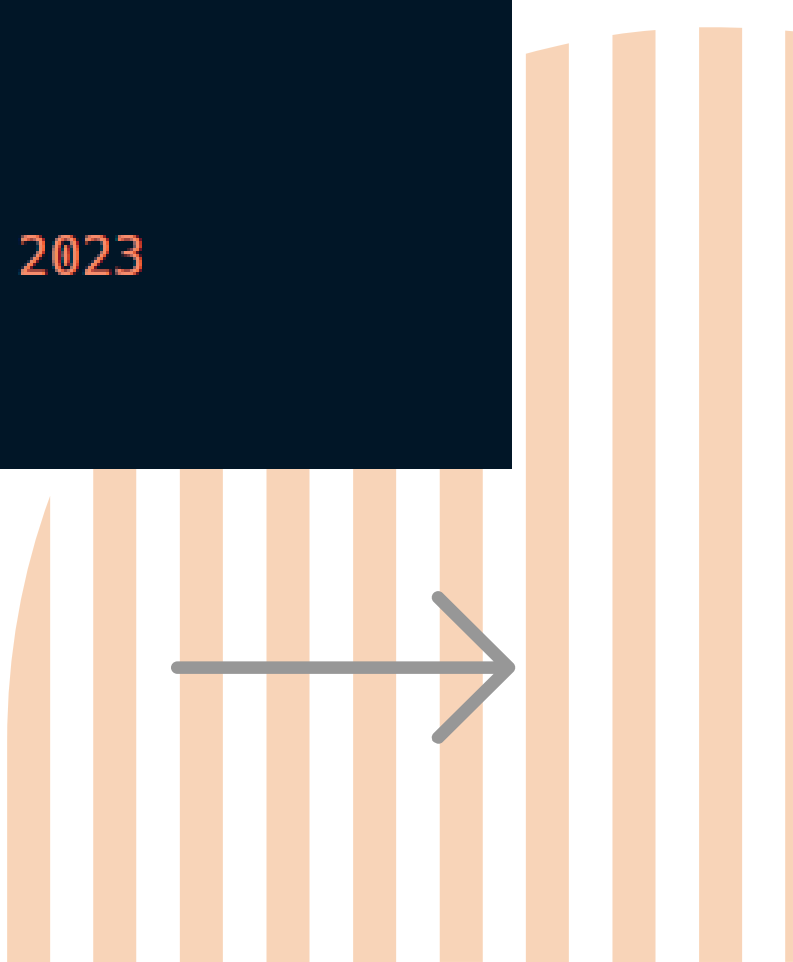


```
CALCULATETABLE(  
    SalesTable,  
    YEAR([Purchase Date]) = 2023  
)
```

Retrieve customers who purchased over \$200 in 2023:

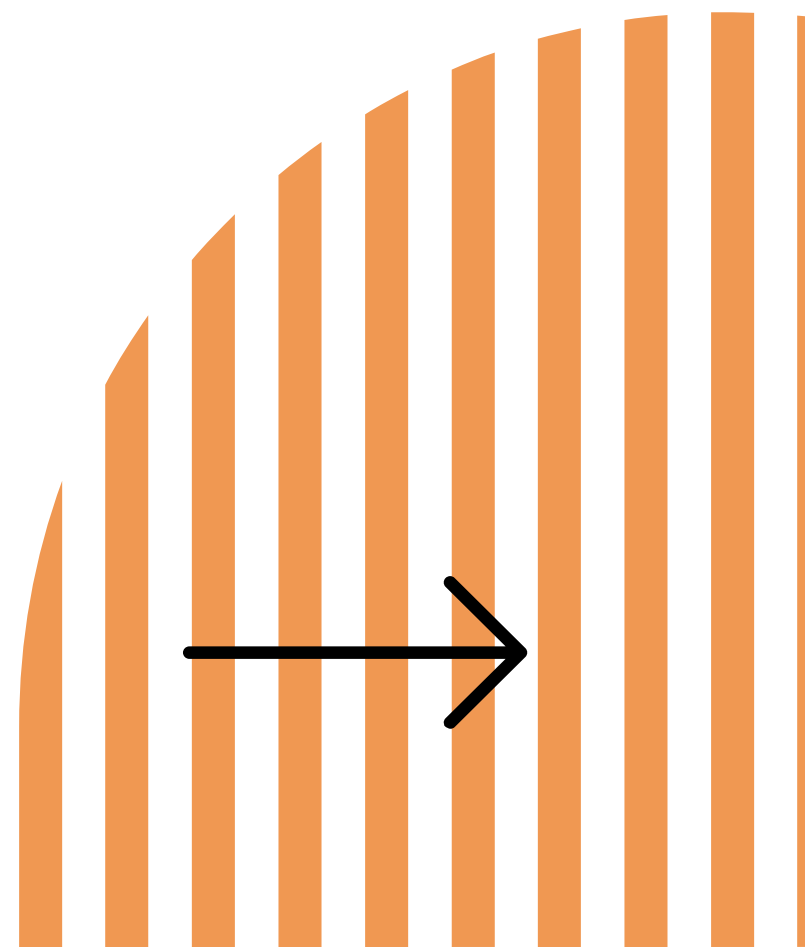


```
CALCULATETABLE(  
    SalesTable,  
    SalesTable[Amount] > 200,  
    YEAR(SalesTable[Purchase Date]) = 2023  
)
```



Additional Tips:

- **Performance:** Use `CALCULATE` for simpler aggregations, while `CALCULATETABLE` is ideal when returning filtered tables for further analysis or visualization.
- **Context Awareness:** Always be mindful of the context in which you're applying these functions, as they can drastically change the results based on the filters applied.



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